Erica L. McCormick

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EDUCATION

Stanford University 2027 (Expected)
Ph.D. in Earth System Science

Stanford University 2026
M.S. in Earth System Science

University of Texas at Austin 2020
B.S. Environmental Science, Geology

AWARDS, HONORS, AND FELLOWSHIPS

External Agencies

Outstanding Student Presentation Award, the American Geophysical Union Hydrology Section, Remote Sensing Technical Committee 2025

Outstanding Student Presentation Award, the American Geophysical Union, Biogeosciences Section 2024

National Science Foundation Graduate Research Fellowship (GRFP) 2022
ORISE Summer Research Fellowship (US Forest Service) 2022
Garden Club of America Fellowship in Urban Forestry 2019

Stanford University

Graduate Community-Engaged Teaching Fellowship, Haas Center 2025
Diversifying Academia, Recruiting Excellence (DARE) Doctoral Fellowship 2024

University of Texas at Austin

UT Honors Program Skaaren Climate Fellowship 2019
UT Honors Program Travel Grant for Undergraduate Research 2018

PUBLICATIONS

Google Scholar (407 citations, H-index: 7), *denotes mentee*

9. **E.L. McCormick**, L.E. Sanders, K.A. McColl, A.G. Konings. (2026). Triple collocation validates CONUS-wide evapotranspiration inferred from atmospheric conditions. *Hydrology and Earth System Sciences*, *30*(8), 2417-2432.

8. **E.L. McCormick**, C. Famiglietti, D. Feng, A.M. Michalak, A.G. Konings (2025). Susceptibility to Photosynthesis Suppression From Extreme Storms Is Highly Site-Dependent. *Global Change Biology*, *31*(5), e70257.

7. M. Zhao, **E.L. McCormick**, G. A, A.G. Konings, B. Li. (2025). Substantial root-zone water storage capacity observed by GRACE and GRACE/FO. *Hydrology and Earth System Sciences*, *29*(10), 2293–2307.

6. A.G. Konings, K. Rao, **E.L. McCormick**, A.T. Trugman, A.P. Williams, N.S. Diffenbaugh, M. Yebra, M. Zhao. (2024). Species cover explains only half of spatial variability in plant water sensitivity. *Global Change Biology*, *30*(7), e17425.

5. W.J. Hahm, D.A. Lapides, D.M. Rempe, **E.L. McCormick**, D.N. Dralle. (2022). The age of evapotranspiration: lower-bound constraints from distributed water fluxes across the continental United States. *Water Resources Research*, *58*(10), e2022WR032961.

4. **E.L. McCormick,** D.N. Dralle, W.J. Hahm, A.K. Tune, L. Schmidt, K.D. Chadwick, D.M. Rempe. (2021). Widespread woody plant use of water stored in bedrock. *Nature*, *597* (7875), 225-229.

Updated 11/01/2026

3. D.N. Dralle, W.J. Hahm, K.D. Chadwick, **E.L. McCormick**, D.M. Rempe. (2021). Accounting for snow in the estimation of root-zone water storage capacity from precipitation and evapotranspiration fluxes. *Hydrology and Earth System Sciences*, *25*(5), 2861-2867.

2. Matheny, A.M., P. Marchetto, J. Powell, A. Rechner, J.Y. Chuah, **E.L. McCormick**, S. Pierce. (2019). LEAF: Logger for Ecological and Atmospheric Factors. *HardwareX*, *6*, e00079.

1. A.R. Mursinna, **E.L. McCormick**, K. Van Horn, L. Sartin, A.M. Matheny. (2018) Plant hydraulic trait covariation: a global meta-analysis to reduce degrees of freedom in trait-based hydrologic models. *Forests*, *9*(8), 446. (Cover Article).

SUBMITTED / IN REVIEW / IN REVISION

3. **E.L. McCormick**, Y.F. Reinfelder, D.M. Rempe, D.A. Lapides, K.A. McColl, A.G. Konings. (In Review). Rock moisture frequently supplies more than half of daily evapotranspiration. *Nature Water.*

2. F. Giardina et al. (including **E.L. McCormick)**. (In Revision). Connecting plant-water interactions across scales: historical discrepancies and emergent convergence. *New Phytologist* (Tansley Review).

1. A.M. Donaldson, D. Smith, L. Schmidt, **E.L. McCormick**, M. Lee, W.J. Hahm, D.N. Dralle, D.A. Lapides, K.D. Chadwick, D.M. Rempe. (Submitted). Memory of multi-year drought recorded in the deep root-zone. *Geophysical Research Letters.*

BOOK CHAPTERS

1. D.M. Rempe, **E.L. McCormick**. (2026). Ch. 9, The Ecohydrology of Roots and Rock. In: B.P. Wilcox, H. Asbjornsen, K. Smettem, I. Creed, L. Wang (eds.), Handbook of Terrestrial Ecohydrology. Routledge. ISBN: 9781032469454.

INVITED TALKS

Quantifying sources of vegetation water use with remote sensing 2026
USDA Southwest Watershed Research Center, Tucson, AZ

Rock moisture frequently contributes to daily ET. 2024
American Geophysical Union Fall Meeting, Washington, D.C.
Won "Outstanding Student Presentation Award" from the Biogeosciences Section

Rock moisture and its implications for ecosystem resilience to precipitation variability 2023
Hydro90 Conference, China (Virtual).

Weathered bedrock commonly supplies water to woody plants. 2021
American Geophysical Union Fall Meeting, New Orleans.

SELECTED CONTRIBUTED PRESENTATIONS

Triple Collocation Validates ET Using the Surface Flux Equilibrium Method. 2025
American Geophysical Union Fall Meeting, New Orleans.
Won "Outstanding Student Presentation Award" from the Hydrology Section, Remote Sensing Technical Committee

2025
How can we quantify sources of water for plants… from space?
Stanford Joint Hydrology Seminar

Rock moisture frequently contributes to daily ET. 2025
Missing links in drought research workshop, Monte Verità, Switzerland.

Updated 11/01/2026

Rock moisture frequently contributes to daily ET. 2024
Computational Methods in Water Research (CMWR), Tucson, Arizona.

Rock moisture frequently contributes to daily ET. 2024
Stanford Joint Hydrology Seminar

Water scarcity and abundance: the effects of drought and extreme wet events on vegetation 2023
Stanford Joint Hydrology Seminar

RESEARCH EXPERIENCE

Graduate Researcher, Stanford University 2022-Present
Remote Sensing Ecohydrology Group, Doerr School of Sustainability. Advisor: Alexandra Konings

- Used satellite remote sensing to show that rock moisture frequently supplies more than half of daily ET across the Western US (submitted, *Nature Water)*
- Created and validated a publicly available ET dataset for the contiguous US by applying a recent theory that infers ET from atmospheric conditions (*HESS*, 2026)
- Applied machine learning to examine the response of photosynthesis to extreme storms (*Global Change Biology*, 2025)
- Supervised three undergraduate researchers, including one as a 2nd author
- Mentored a graduate student on machine learning analyses of fuel treatment impacts on vegetation using SFE ET (*Hess, 2026)* product as target

ORISE Summer Research Fellow, US Forest Service Sum 2022
Supervisor: David Dralle

- Investigated how hydrologic factors shape visitation to California's National Forests, combining visitor cell-phone data with satellite remote sensing

Research Engineering/Scientist Assistant (Full-Time), University of Texas at Austin 2020-2022
Vadose Zone Ecohydrology Lab, Jackson School of Geosciences. Advisor: Daniella Rempe

- Created maps of bedrock water storage across the US using remote sensing (*Nature,* 2021)
- Measured rock moisture in the field using geophysical methods (nuclear magnetic resonance and neutron probe)
- Helped design a lab experiment testing nuclear magnetic resonance for measuring bedrock fracture size
- Contributed to grant proposals and science communication articles
- Developed WaterPyk, a Python package for analyzing hydrological data with Google Earth Engine, and taught it to two lab groups through a workshop and video tutorial

Undergraduate Researcher, University of Texas at Austin 2018-2020
Ecohydrology Lab, Jackson School of Geosciences. Advisor: Ashley Matheny

- Conducted global meta-analysis of how plant hydraulic traits co-vary (*Forests*, 2018)
- Designed, built, and deployed sensors to monitor urban vegetation and precipitation (*HardwareX*, 2019)
- Applied plant hydraulics model (FETCH) to new vegetation types
- Helped build and deploy sap flow sensors at the U of Michigan Biological Station

Undergraduate Researcher, University of Texas at Austin 2017-2020
Geoarchaeology and Soil Lab, Department of Geography. Advisors: Timothy Beach and Sara Eshleman

- Designed and led a project comparing soil carbon measurement techniques on understudied lowland and highland soils in Belize and Peru
- Conducted a month of fieldwork in remote Belizean rainforest, including exploratory caving, soil sampling, and archaeological and vegetation mapping
- Performed soil chemical analyses (carbon, nitrogen, phosphorus, grain size, magnetic susceptibility)

MENTORSHIP

Pavan Siddharth Kosuru, Undergraduate, Las Positas College (now at UC Berkeley) 2025-Present

- *Project: Comparing the response of photosynthesis and evapotranspiration to extreme storms using machine learning.*
- Recruited through Stanford Science Small Groups, a mentorship program for community college students that I volunteer with

Lillian Sanders, Undergraduate, Stanford University 2023-2025

- *Project: Surface flux equilibrium ET implementation and validation with triple collocation*
- 2nd author, McCormick et al. (2026)
- Full-time summer research intern in 2024
- Two poster presentations and one eLightning Oral session at AGU 2025

Bhu Kongtaveelert, Undergraduate, Stanford University

- *Project: Effects of rice field drainage timing on methane emissions using remote sensing* Sum 2023
- Summer research intern, co-supervised with Jack Lamb (Stanford PhD student)
- Poster presentation, AGU 2023

TEACHING EXPERIENCE

Stanford University

Stanford Graduate Summer Institute, "Jumpstart Your Academic Job Search" Sums 2025, 2026

- Enrollment: 67, mix of graduate students and postdocs
- Assisted with course design, organization, and guest speakers and panelists
- Taught session on how to brainstorm questions to ask during campus visits

ESS 223 Biosphere-Atmosphere Interactions Win 2023

- Enrollment: 27, mix of graduate and undergraduate students.
- Held weekly office hours and assisted with grading of 5 problem sets

University of Texas at Austin

GEO 391 Vadose Zone Hydrology Spr 2021

- Enrollment: 18 graduate and undergraduate students
- Prepared and gave lecture on Hydrus 1D (soil modeling) software
- Prepared and gave lecture on Google Earth Engine
- Assisted with grading and redesigning of 6 assignments

TC 358 Law and Ethics of Climate Change Fall 2020

- Enrollment: 18 undergraduate students
- Prepared and gave lecture on hydrogeology issues of central Texas
- Assisted with assignment development, organization, and guest speakers

SERVICE

Service to the University

Department Wellness Liaison. Relayed student needs to departmental leaders and administration; planned events to link students to campus mental health resources.	2024-2025
Residential Community Associate, Munger Residences, Stanford Graduate Housing	2025
Student Committee Member for Faculty Search in Earth System Science Department	2024

Service to the Discipline

Proposed and co-convened AGU session "Rock Moisture in Weathered Bedrock: Understanding the Hydrologic, Ecological, and Geochemical Function of Water Movement and Storage Within Weathered Bedrock"	2025
Invited Panelist, "Developing Effective Mentoring Relationships" webinar, Consortium of Universities for the Advancement of Hydrologic Science	2023
Invited Panelist, "Professional Development for Environmental Scientists" course, University of Texas at Austin	2021-2022
Reviewer for *Nature Geoscience*, *Nature Communications*, *Geophysical Research Letters*, *AGU Advances*, *Journal of Hydrology*, *Earth System Dynamics*, *Water Resources Research*	2020-Present

Outreach

Guest lecture for IB Environmental Science course at Redwood City High School	2025
Mentor for Stanford Science Small Groups community college outreach program	2024-2025
Volunteer Pen-Pal, Letters to a Pre-Scientist organization	2022-2025

PRESS

Scientific American 2021
[link] "Trees Drill Into Deep Bedrock for Water Surprisingly Often" by Tess Joosse

Science & Vie Magazine, France 2021
"Les Arbres Boivent De L'eau Dans Les Roches"

Eos Magazine 2021
[link] "Thirsty Plants Pull Water from Bedrock" by Katherine Kornei

University of Texas News 2021
[link] "Water in Bedrock is Sustaining Trees Across the Country" by Monica Kortsha

Simon Fraser University News 2021
[link] "Could the Water in Bedrock Save our Forest Ecosystems from Climate Change?"

SOFTWARE & DATA

denotes mentee

E.L. McCormick, L.E. Sanders, K.A. McColl, A.G. Konings. (2026). Daily surface flux equilibrium (SFE) evapotranspiration across CONUS [Data set]. Zenodo. https://doi.org/10.5281/zenodo.21049346.
561 downloads.

E.L. McCormick, D.N. Dralle, W.J. Hahm, A.K. Tune, L. Schmidt, K.D. Chadwick, D.M. Rempe. (2021). Dataset for "Evidence for widespread woody plant use of water stored in bedrock" [Data set]. HydroShare. https://doi.org/10.4211/hs.a2f0d5fd10f14cd189a3465f72cba6f3.
2,088 downloads.

E.L. McCormick. (2022). "WaterPyk," a Python package for downloading and analyzing hydrological timeseries at the site, polygon, or USGS watershed level. Leverages Google Earth Engine cloud computing platform. https://github.com/erica-mccormick/waterpyk.
27,055 downloads (PyPI).

SOCIETAL AFFILIATIONS

American Geophysical Union

Updated 11/01/2026